

Duality III: Prices and Certificates

The same constraint weights can explain the value of a requirement, the cost of a column, and why a system has no feasible solution.

Our route: numerical prices $\rightarrow$ column values and geometry $\rightarrow$ an infeasibility certificate.

Bertsimas and Tsitsiklis §§4.4 and 4.6.

Recall the example

Interpret a price

Value a column

Prove infeasibility

Optional derivations

The quantities and their costs are already familiar

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

The optimal quantities are $x^* = (1, 0, 1)$, with cost 19.

The two equality right-hand sides are requirements that must be matched exactly.

Now ask: what happens to the best cost when one requirement changes?

The optimal weights are a certificate we can reuse

$$\begin{array}{ll}\max & 8p_1 + 3p_2 \\ \text{s.t.} & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \text{ unrestricted.}\end{array}$$

$$p^* = (2, 1), \quad 8(2) + 3(1) = 19.$$

The identity $13x_1 + 10x_2 + 6x_3 = 2(5x_1 + x_2 + 3x_3) + (3x_1 + x_2) + 7x_2$ proves the matching bound.

Increase the first requirement from 8 to 9

Keep $x_2 = 0$, and solve the changed equations:

$$3x_1 = 3 \Rightarrow x_1 = 1, \quad 5(1) + 3x_3 = 9 \Rightarrow x_3 = \frac{4}{3}.$$

$$13(1) + 6\left(\frac{4}{3}\right) = 21 = 19 + 2(1).$$

The same weights give the lower bound $2(9) + 1(3) = 21$. This feasible solution attains it, so the cost really increases by 2.

The first price $p_1 = 2$ measures cost per unit increase in requirement 1.

Increase the second requirement from 3 to 3.6

Keep the first requirement at 8 and again set $x_2 = 0$:

$$\begin{aligned} 3x_1 &= 3.6 \Rightarrow x_1 = \frac{6}{5}, \\ 5\left(\frac{6}{5}\right) + 3x_3 &= 8 \Rightarrow x_3 = \frac{2}{3}. \end{aligned}$$

$$13\left(\frac{6}{5}\right) + 6\left(\frac{2}{3}\right) = 19.6 = 19 + 1(0.6).$$

The dual lower bound is also $2(8) + 1(3.6) = 19.6$. Thus $p_2 = 1$ is the cost per extra unit of requirement 2 here.

A price prediction is exact while the basis stays feasible

If the right-hand side changes from b to $b + \Delta b$, and the same optimal basis stays feasible, then

$$v(b + \Delta b) = v(b) + (p^*)^\top \Delta b.$$

Changing only the right-hand side leaves the dual constraints and the reduced costs unchanged.

The original dual weights still give a valid lower bound after any right-hand-side change. Equality with the new optimum needs a feasible primal solution that attains that bound.

For a nondegenerate optimal basis, sufficiently small changes in any direction preserve feasibility.

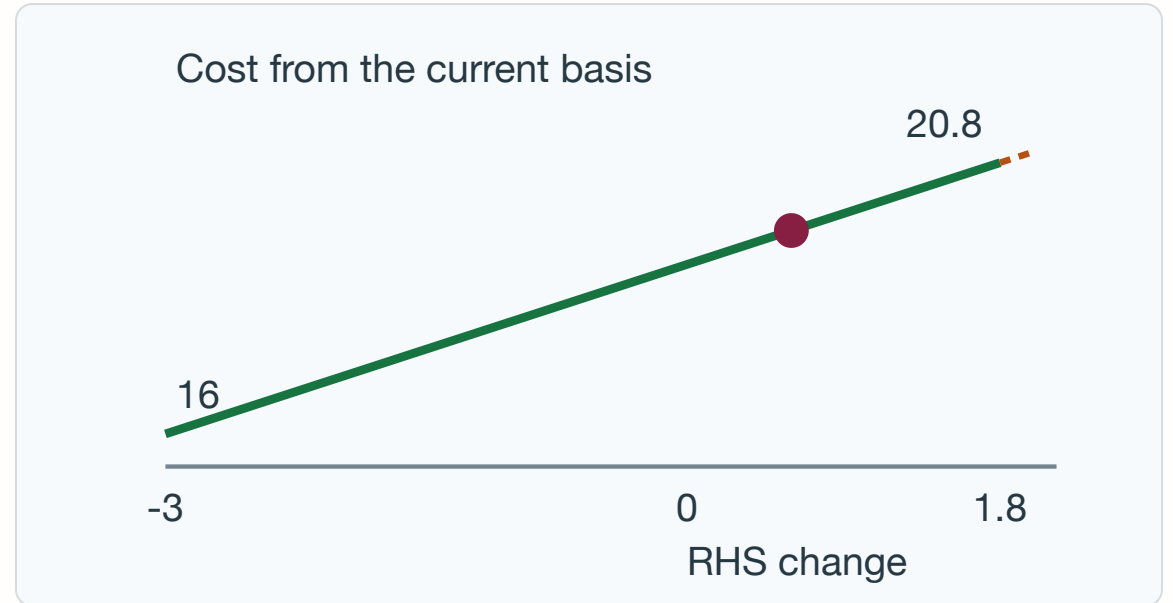
How far can we keep the same quantities pattern?

Replace 3 by $3 + \varepsilon$ in the second equality, keeping $x_2 = 0$.

$$x_1 = 1 + \frac{\varepsilon}{3}, \quad x_3 = 1 - \frac{5\varepsilon}{9}.$$

$$v = 19 + \varepsilon \quad \text{while} \quad -3 \leq \varepsilon \leq \frac{9}{5}.$$

Shown: a change of 0.6 keeps both basic quantities positive.



Change 0.6: quantities (1.2, 0, 0.667); cost 19.6.
The current basis remains feasible and optimal.

A negative quantity ends this basis's prediction

At $\varepsilon = 2$, the second requirement is 5. The old pattern gives

$$x_1 = \frac{5}{3}, \quad x_2 = 0, \quad x_3 = -\frac{1}{9}.$$

This is infeasible. Its predicted cost 21 is not the cost of a feasible solution.

The weights $(2, 1)$ still certify the lower bound $2(8) + 5 = 21$. We must reoptimize to find the actual minimum.

Optional: solve the new optimum explicitly.

Interpret the price with the constraint in view

Equality requirements: a price is the change in minimum cost per unit increase in a required total. Equality multipliers can be positive, zero, or negative.

Range: the prediction holds while a suitable optimal basis remains feasible. At a change of basis, the marginal cost can change.

At degeneracy, prices may not be unique and the value function may have a corner. A dual multiplier still gives a valid bound.

We will calculate allowable changes systematically in sensitivity analysis.

Interpret the columns as ingredients and the weights as prices

Think of each column as the contents of one unit of an ingredient. The two equality rows specify the required contents of the final mixture.

Ingredient	Contents A_j	Cost c_j	Value $(p^*)^\top A_j$
1	(5, 3)	13	$2(5) + 1(3) = 13$
2	(1, 1)	10	$2(1) + 1(1) = 3$
3	(3, 0)	6	$2(3) + 1(0) = 6$

Dual feasibility says the value assigned to an ingredient's contents cannot exceed its purchase cost.

Used ingredients are priced exactly at their cost

$$c - (A^\top p^*) = (0, 7, 0), \quad x^* = (1, 0, 1).$$

Ingredients 1 and 3 are used, and their price–cost gaps are zero. Ingredient 2 costs 7 more than the value assigned to its contents, so it is unused.

$$13x_1 + 10x_2 + 6x_3 = 19 + 7x_2.$$

This is the dictionary objective from simplex. Its reduced cost 7 measures the extra cost of using ingredient 2 after adjusting the basic ingredients.

A zero gap permits positive use; it does not force it. The whole optimal mixture is also priced exactly: $(p^*)^\top b = 19$.

The mixture problem behind the picture

Column	B	C	D	E	F
First total	1/2	3	0	2	0
Second total	1/2	0	3	1/2	0
Cost	7	4	4	1	4

$$\begin{aligned}
 \min \quad & 7x_B + 4x_C + 4x_D + x_E + 4x_F \\
 \text{s.t.} \quad & \frac{1}{2}x_B + 3x_C + 2x_E = 1, \\
 & \frac{1}{2}x_B + 3x_D + \frac{1}{2}x_E = 1, \\
 & x_B + x_C + x_D + x_E + x_F = 1, \quad x \geq 0.
 \end{aligned}$$

Each x_j is a mixture weight. The two required totals are $(1, 1)$; the third equality makes the weights sum to 1. Thus a dual bound uses **three** row weights.

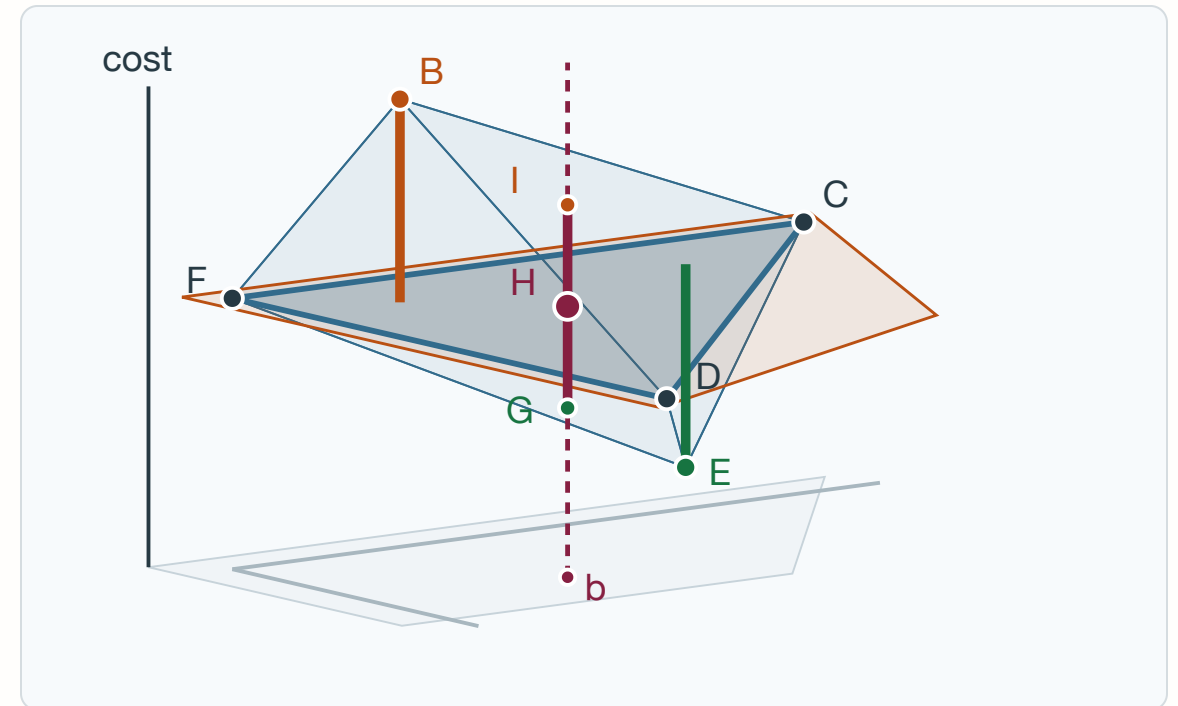
Matching three columns does not yet give a bound

Set $x_C = x_D = x_F = 1/3$, with the others zero.
The totals are $(1, 1)$ and the cost is 4.

Their plane $z = 4$ uses row weights $(0, 0, 4)$: zero on the first two requirements, 4 on the sum-to-one row.

E costs $1 < 4$, so this plane fails its dual inequality. Its gap is $1 - 4 = -3$.

Matching C, D, F alone does not bound every available mixture.



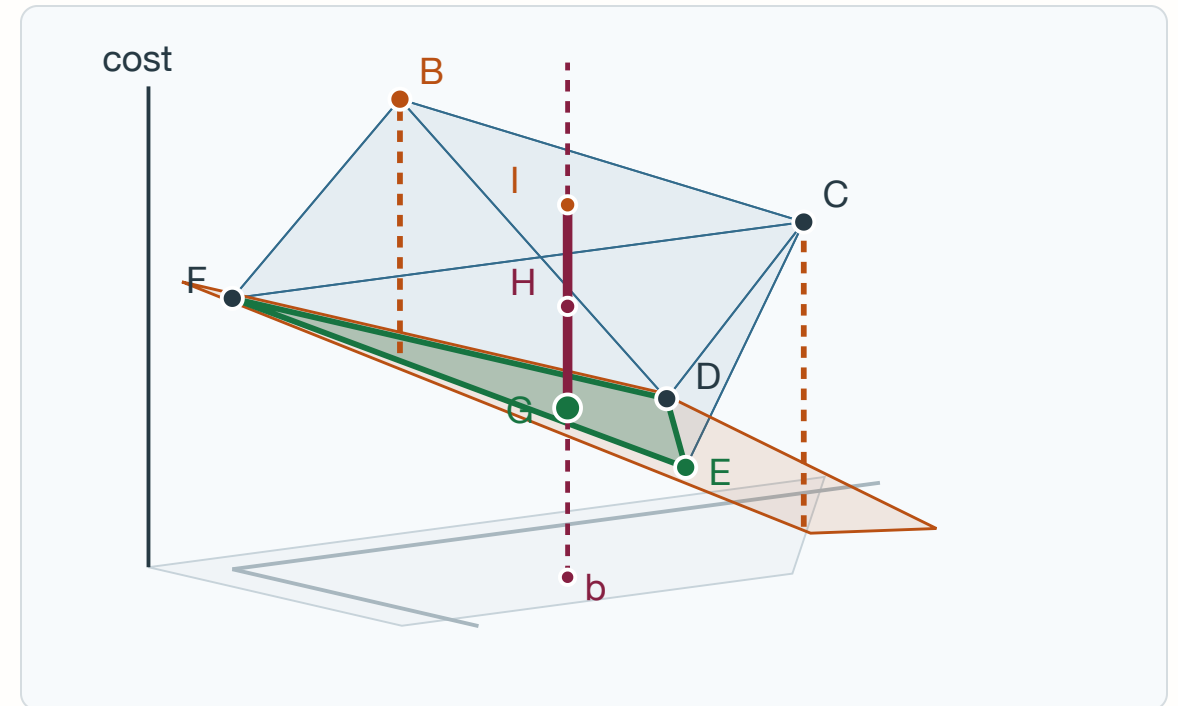
Weight the three requirements to get a lower bound

Use $p = (-3/2, 0, 4)$. For a column with entries (a_1, a_2) , its weighted value is the plane height $z = 4 - \frac{3}{2}a_1$.

The constant 4 weights the sum-to-one equality.

Every column lies on or above this plane. At the required totals, it gives the bound $-\frac{3}{2}(1) + 0(1) + 4(1) = \frac{5}{2}$.

The mixture $(x_D, x_E, x_F) = (1/4, 1/2, 1/4)$, with the others zero, attains $5/2$.



Weighted column bounds certify the mixture

$A_j = (a_{1j}, a_{2j}, 1)$ includes normalization. Use $p = (-3/2, 0, 4)$.

$$c_j \geq p^\top A_j = -\frac{3}{2}a_{1j} + 0a_{2j} + 4 \quad \text{for every column } j.$$

Multiply by $x_j \geq 0$, add, and use the three required totals:

$$\begin{aligned} c^\top x &\geq -\frac{3}{2} \sum_j a_{1j}x_j + 0 \sum_j a_{2j}x_j + 4 \sum_j x_j \\ &= -\frac{3}{2}(1) + 0(1) + 4(1) = \frac{5}{2}. \end{aligned}$$

D, E, F attain the bound. B and C have dual slacks (reduced costs) $15/4$ and $9/2$.

Can the same columns meet a different target?

Keep the columns $(5, 3)$, $(1, 1)$, $(3, 0)$, but change the target from $(8, 3)$ to $(1, 2)$:

$$5x_1 + x_2 + 3x_3 = 1, \quad 3x_1 + x_2 = 2, \quad x \geq 0.$$

We are asking only whether any nonnegative quantities satisfy these equations. No objective is needed.

Instead of trying many quantities, look for a combination of the equations that is impossible.

Subtract the second equation from the first

$$5x_1 + x_2 + 3x_3 = 1, \quad 3x_1 + x_2 = 2, \quad x \geq 0.$$

$$(5x_1 + x_2 + 3x_3) - (3x_1 + x_2) = 1 - 2,$$

$$2x_1 + 3x_3 = -1.$$

The left side is nonnegative whenever the quantities are nonnegative. It cannot equal -1 .

The row weights $q = (1, -1)$ give a short, checkable **certificate of infeasibility**.

Farkas' lemma: exactly one alternative holds

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Alternative 1 — feasibility: there is a nonnegative solution.

$$Ax = b, \quad x \geq 0.$$

Alternative 2 — infeasibility: there are row weights satisfying

$$A^\top q \geq 0, \quad b^\top q < 0.$$

Theorem 4.6 says exactly one alternative holds. The weights q are unrestricted; unlike quantities, they need not be nonnegative.

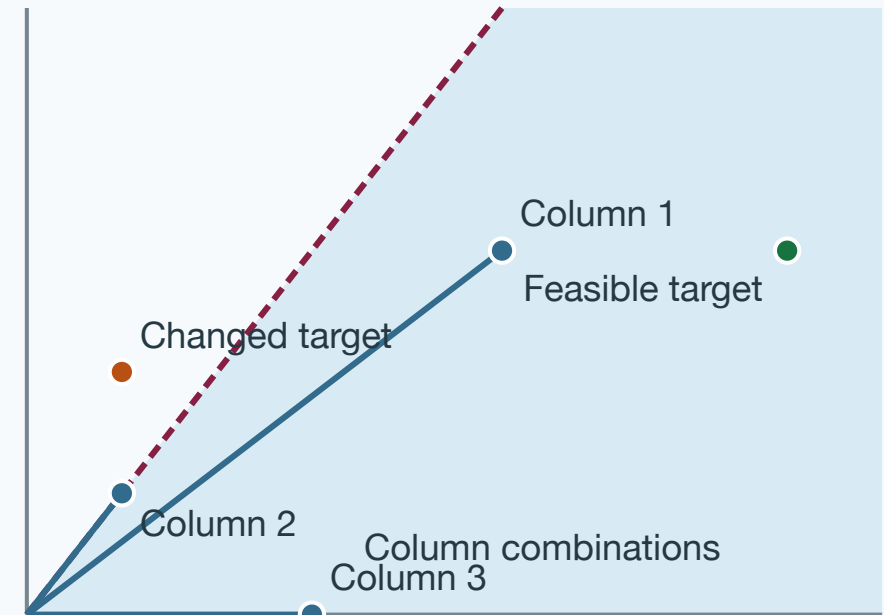
A certificate separates the target from the column cone

Every attainable target is $Ax = x_1A_1 + x_2A_2 + x_3A_3$ with $x \geq 0$. These combinations form a **cone**.

For $q = (1, -1)$, all columns satisfy $q^\top A_j \geq 0$. The changed target satisfies $q^\top (1, 2) = -1$.

There is no sum-to-one equation here. The coefficients need only be nonnegative, so the picture is a cone rather than a convex hull.

Equation 2 total



Equation 1 total

The dashed separator is $a_1 - a_2 = 0$. Every column has $a_1 - a_2 \geq 0$; the changed target has $1 - 2 = -1$.

Why the certificate rules out every feasible point

Suppose $A^\top q \geq 0$ and $b^\top q < 0$. If $Ax = b$ and $x \geq 0$ also held, then

$$0 \leq (A^\top q)^\top x = q^\top Ax = q^\top b < 0.$$

This contradiction proves that a proposed certificate is sufficient. Farkas' lemma also guarantees that one exists whenever the system is infeasible.

The existence proof using Phase I and strong duality is in the appendix.

Check the certificate without solving the LP

Use the same columns and $q = (1, -1)$. First compute $A^\top q$, then check the target $b = (2, 3)$.

$$A^\top q = (2, 0, 3) \geq 0, \quad b^\top q = 2 - 3 = -1 < 0.$$

The weights certify infeasibility. For the old target (8,3), the weighted right-hand side is 5, so these weights do not certify infeasibility.

Three uses of the same row-combination idea

Question	What we check
Is this feasible solution optimal?	A feasible dual bound equals its cost.
What is one extra unit worth?	The optimal multiplier, within its validity range.
Is this equality system infeasible?	A row combination gives a nonnegative left side and a negative right side.

Next: calculate sensitivity ranges, then use dual prices to decide which new columns could improve a restricted LP.

The optimization problems remain continuous linear programs.

Study guide

Be able to: verify a price by solving a small changed problem; distinguish a bound from an attainable value; compute a column's price–cost gap; verify an infeasibility certificate.

Read textbook §4.4 for marginal costs and §4.6 for Farkas' lemma. The matrix derivation and existence proof below are optional extensions.

Calculate the price range

Recover the matrix formula

Connect Farkas to Phase I

Appendix: derivations and further examples

These pages supply the full calculations behind the main examples and the existence part of Farkas' lemma.

Allowable RHS range

Reoptimize beyond the range

Matrix formula for prices

Existence via Phase I

An inequality-system alternative

Calculate the range from nonnegative quantities

Keep $x_2 = 0$, the first target 8, and the second target $3 + \varepsilon$.

$$x_1 = 1 + \frac{\varepsilon}{3} \geq 0 \iff \varepsilon \geq -3,$$

$$x_3 = 1 - \frac{5\varepsilon}{9} \geq 0 \iff \varepsilon \leq \frac{9}{5}.$$

Thus the current basis is feasible for $-3 \leq \varepsilon \leq 9/5$. The reduced costs stay $(0, 7, 0)$, so this basis stays optimal throughout the interval.

At an endpoint a basic quantity is zero. At an interior point both basic quantities are positive.

Beyond the range, a new basis gives a new price

With target $(8, 5)$, set $x_3 = 0$ and solve

$$5x_1 + x_2 = 8, \quad 3x_1 + x_2 = 5 \quad \Rightarrow \quad x = (3/2, 1/2, 0).$$

$$c^\top x = \frac{49}{2} = 24.5.$$

Tight dual constraints for the positive variables give $5p_1 + 3p_2 = 13$ and $p_1 + p_2 = 10$, hence $p = (-17/2, 37/2)$.

The remaining inequality is $3p_1 = -51/2 \leq 6$; the dual value is $8(-17/2) + 5(37/2) = 49/2$. This certifies the new optimum.

The arithmetic has a compact matrix form

For an optimal basis B with $B^{-1}b \geq 0$ and nonnegative reduced costs, the quantities and weights are

$$x_{\mathcal{B}} = B^{-1}b, \quad (p^*)^\top = c_{\mathcal{B}}^\top B^{-1}.$$

$$\begin{aligned} c_{\mathcal{B}}^\top B^{-1}(b + \Delta b) &= c_{\mathcal{B}}^\top B^{-1}b + c_{\mathcal{B}}^\top B^{-1}\Delta b \\ &= v(b) + (p^*)^\top \Delta b. \end{aligned}$$

This is the optimum whenever $B^{-1}(b + \Delta b) \geq 0$, because the reduced costs do not depend on b .

Why nondegeneracy gives room for small changes

Assume $B^{-1}b > 0$. Every current basic quantity is strictly positive.

The changed quantities are $B^{-1}b + B^{-1}\Delta b$. For sufficiently small Δb , none can reach zero.

Thus the same basis stays feasible in a neighborhood, and the value is a linear function there. Its coefficient on requirement i is the optimal multiplier for that row.

If a basic quantity already equals zero, some arbitrarily small changes may make it negative. This explains why the nondegeneracy assumption matters for a two-sided neighborhood.

Existence of a certificate: start with Phase I

First multiply any equality with a negative right-hand side by -1 , so the transformed system has $b \geq 0$.

$$\min \mathbf{1}^\top y \quad \text{s.t. } Ax + y = b, \quad x, y \geq 0.$$

This auxiliary LP is feasible at $x = 0, y = b$ and bounded below by zero. It attains a finite optimum w^* .

If the original system is infeasible, $w^* > 0$: an optimum of zero would force every artificial variable to zero and give $Ax = b$.

Strong duality supplies the separating weights

The dual of the Phase I problem is

$$\max b^\top p \quad \text{s.t.} \quad A^\top p \leq 0, \quad p \leq \mathbf{1}, \quad p \text{ unrestricted.}$$

Strong duality gives an optimal p^* with $b^\top p^* = w^* > 0$. Set $q = -p^*$.

$$A^\top q \geq 0, \quad b^\top q = -w^* < 0.$$

This proves existence when the original system is infeasible. The earlier contradiction proves that the two alternatives cannot both hold.

Return the certificate to the original row signs

Write the row sign changes as D , a diagonal matrix with entries 1 or -1 . The transformed system is $(DA)x = Db$.

Suppose its certificate is $\hat{q}$. Set $q = D\hat{q}$. Since $D^\top = D$,

$$A^\top q = (DA)^\top \hat{q} \geq 0, \quad b^\top q = (Db)^\top \hat{q} < 0.$$

Thus the certificate applies to the original equations. No sign assumption on the original right-hand side is needed for Farkas' lemma.

Another alternative: inequalities and free variables

For $Ax \leq b$ with x unrestricted, exactly one of the following holds:

There exists x with $Ax \leq b$.

There exists $q \geq 0$ with $A^\top q = 0$, $b^\top q < 0$.

To derive it, write $x = u - v$, $u, v \geq 0$, and add slack $s \geq 0$. Apply the equality-form lemma to $[A, -A, I](u, v, s)^\top = b$.

The column conditions are $A^\top q \geq 0$, $-A^\top q \geq 0$, and $q \geq 0$. These simplify to the stated conditions.